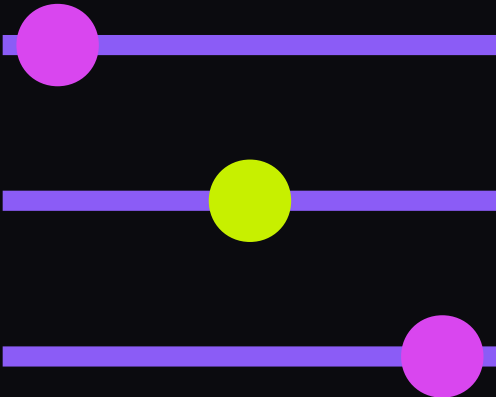


BLUEMOON BROWSER- CURL CHAIN

Detects curl.exe writing beneath Temp when launched by cmd.exe whose parent is a Chromium-family browser—the distinctive process tree reported for default BlueMoon exploitation.

PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM



DeviceProcessEvents

Behavioral with campaign context

EVIDENCE BEFORE ALERTING

REPORT DATE	STATUS
13 September 2026	Untested implementation sketch
PRIMARY PLATFORM	OPERATIONAL VALUE
Microsoft Defender XDR	9.8 / 10



EXECUTIVE ASSESSMENT

Decision, confidence and evidence boundary.

ASSESSMENT

OPERATIONAL VALUE

9.8

Detects curl.exe writing beneath Temp when launched by cmd.exe whose parent is a Chromium-family browser—the distinctive process tree reported for default BlueMoon exploitation.

STATUS

Production candidate

Production candidate - Untested implementation sketch; Confidence Medium.

- 1primary table
- 0thresholds
- 0joins
- 0secondary rules

Evidence boundary

A result does not prove exploitation, payload execution, actor attribution, or compromise. The query does not inspect browser memory or web content.



BEHAVIOR AND EVIDENCE

Observed facts separated from analytical scope.

MECHANICS

Source-supported fact

- 1 Proofpoint directly observed BlueMoon produce chrome.exe → cmd.exe → curl.exe → payload execution.
- 2 The default command uses curl with -o to write an executable under the current user's Temp directory.
- 3 The same exploit kit was adopted by four espionage-motivated clusters from 28 August into September 2026.

Detection hypothesis

If an attacker repeats this behavior, process telemetry should show the source-observed command and ancestry. It can be distinguished through the exact execution context plus provenance and adjacent host evidence.

Unconfirmed elements

A result does not prove exploitation, payload execution, actor attribution, or compromise. The query does not inspect browser memory or web content.

Logic: Behavioral with campaign context **ATT&CK:** T1059.003 · T1105



TELEMETRY CONTRACT

Collection prerequisites and documented fields.

TELEMETRY

REQUIRED

Microsoft Defender for Endpoint process creation telemetry with complete parent and grandparent attribution.

OPTIONAL ENRICHMENT

Browser version/build, URL/domain evidence, file events for the output, payload hash/signature, email delivery and user context.

Fields	Reliability	Use
Timestamp, DeviceId, DeviceName	Documented	Required for query or triage
FileName, FolderPath	Documented	Required for query or triage
ProcessCommandLine	Documented	Required for query or triage
InitiatingProcessFileName	Documented	Required for query or triage
InitiatingProcessParentFileName	Documented	Required for query or triage
ProcessUniqueld, ReportId	Documented	Required for query or triage

Unavailable / excluded

No browser payload, encrypted content, inferred success, rarity score, arbitrary threshold, or undocumented field is used.



PRIMARY KQL

Direct, source-supported conditions with investigable output.

DETECTION

STATUS

Production candidate - Untested implementation sketch

KQL – UNTESTED

```
DeviceProcessEvents
| where FileName =~ "curl.exe"
| where InitiatingProcessFileName =~ "cmd.exe"
| where InitiatingProcessParentFileName in~
  ("chrome.exe", "msedge.exe", "brave.exe", "vivaldi.exe")
| where ProcessCommandLine contains "-o"
| where ProcessCommandLine contains @"\\Temp\\"
| project Timestamp, DeviceId, DeviceName, AccountDomain,
  AccountName, FileName, FolderPath, ProcessCommandLine,
  ProcessId, ProcessUniqueId, InitiatingProcessFileName,
  InitiatingProcessCommandLine,
  InitiatingProcessParentFileName, ReportId
| order by Timestamp desc
```

THRESHOLD	WINDOW	JOIN	AGGREGATION	EXECUTION
None	None	None	None	Untested



TRIAGE AND TUNING

Human validation before response.

WORKFLOW

1

Preserve raw event

Retain projected IDs and complete process context.

2

Confirm provenance

Identify user, maintenance path and initiating application.

3

Enrich

Browser version/build, URL/domain evidence, file events for the output, payload hash/signature, email delivery and user context.

4

Scope

Review adjacent endpoint, identity and network evidence.

Expected benign overlap

Legitimate browser-driven protocol handlers, development/test harnesses, support workflows, or automation that intentionally invokes cmd.exe and curl.exe. Validate browser provenance, download destination, signer/hash, initiating user and adjacent file/network evidence.

Response boundary

A result does not prove exploitation, payload execution, actor attribution, or compromise. The query does not inspect browser memory or web content.



VALIDATION GATES

Controlled tests and cross-SIEM limits.

QUALITY CONTROL

Validation steps

- 1. Run a safe lab chain with a benign local URL and confirm exact grandparent attribution.
- 2. Verify Chrome and Edge separately and confirm case/path normalization.
- 3. Search 30 days to measure benign overlap without adding a rarity threshold.
- 4. Confirm the output file and any subsequent execution through ProcessUniqueld/file pivots.

Decision criteria

Success: The controlled browser → cmd → curl chain matches with full command line and stable ancestry; historical matches are low-volume and explainable.

Rejection: Grandparent attribution is inconsistent, Temp matching is unreliable, or legitimate enterprise workflows dominate results.

Failure: MDE is absent, DeviceProcessEvents lacks ancestry, or command lines are truncated.

Platform	Decision	Missing evidence
Defender XDR	Production candidate	Target-environment execution and baseline
Splunk	No reliable implementation	Index, sourcetype, extraction, sample event
Chronicle	No reliable implementation	Parser and exact UDM mapping



CONCLUSION AND REFERENCES

Coverage statement and document control.

FINAL

CONCLUSION

Production candidate

Detects curl.exe writing beneath Temp when launched by cmd.exe whose parent is a Chromium-family browser—the distinctive process tree reported for default BlueMoon exploitation.

A result does not prove exploitation, payload execution, actor attribution, or compromise. The query does not inspect browser memory or web content.

MITRE ATT&CK

T1059.003 · T1105

References

1. Proofpoint Threat Research — Once in a BlueMoon (09 Sep 2026)
2. Microsoft Learn — DeviceProcessEvents (updated 07 Aug 2026)
3. MITRE ATT&CK — T1059.003 and T1105

Document control

English dossier. No customer data. URLs are defanged in references.txt. HTML and rendering assets remain private.